Statistics 5810/6810: Resampling

Monte Carlo, Bootstrap, Cross Validation

Resampling is a broad term that refers to techniques which use repeated samples from some distribution. Each sample, treated as a pseudo-data set, is analyzed in some way (perhaps just summarized) and by repeating this process, getting slightly different results each time, we can get some idea of the variability in whatever we're doing. We'll look at two methods: bootstrap and cross validation.

Monte Carlo

Monte Carlo Methods are not technically resampling, but they are related and we've done them. In one HW, we sampled repeatedly from the uniform distribution and used these samples to approximate the distribution of the maximum difference between observed quantiles and their expected values. We then compared this approximate distribution to actual maximum differences from the web cache data to decide whether the website update times seemed to come from the uniform distribution. This is an example of a Monte Carlo technique, characterized by taking multiple samples from a known (or presumed) distribution.

An Example

Suppose we have height data on 10 students chosen at random from the class, measured in inches:

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heights = c(66,68,62,69,65,62,70,75,63,61)
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Based on these heights, we can estimate the average height of all the students in the class. Our best guess is the average of the sample heights. How accurate is this guess? One way to answer this question, which you might learn in another statistics class, is to make assumptions about the distribution of the sample average (perhaps it follows the normal curve) and do some calculations. The bootstrap method provides a way to approximate the distribution of the sample average without making any such assumptions.

The Bootstrap Idea

The original sample approximates the population. If we replicate sample values to build up an approximate population, we can then sample repeatedly from this population. In the height example, we then get many samples of size 10 and treat each as a new pseudo-data set and the average is computed. The distribution of the sample averages now gives some idea of what the sample average could have been if a different set of 10 students were chosen. Let's try it in R.

Cross Validation

Cross validation is a technique for assessing how well the results of some statistical analysis will generalize to an independent data set. Using the same data to create the model as well as assess the model can lead to an overly optimistic view of how well the model will be able to make predictions in an independent data set. The idea of cross validation is to leave some data aside as a test set (to assess the predictions)while the rest of the data is a training set used to make decisions about how to make predictions.